

14.7 Answers

① a) $f(1,1) = f(-1,-1) = -1$ is a local min.

$(0,0,1)$ is a saddle point.

b) $f(2,0) = e^4$ local max

c) $f(0,0) = 0$ local min, $f(0, -\frac{5}{3}) = \frac{125}{27}$ local max
 $(\pm 2, -1, 3)$ saddle pts.

d) For every integer k , $(k\pi, 0, 0)$ is a saddle point,

e) $f(1,1) = f(-1,-1) = 4$ local min

② a) $f(0,2) = -9$ abs min
 $f(3,0) = 10$ abs max

b) $f(0,0) = 6$ abs min

$f(-\frac{1}{2}, \pm 1) = \frac{27}{4}$ abs max

c) $f(1,1) = 1$ abs min
 $f(3,0) = 84$ abs max

d) $f(0,y) = f(x,0) = 0$ abs min where
 $0 \leq x, y \leq 2.$

$f(\frac{\sqrt{8}}{\sqrt{3}}, \frac{2}{\sqrt{3}}) = \frac{16}{3\sqrt{3}}$ abs max

③ $(\frac{10}{3}, \frac{7}{3}, \frac{5}{3})$

⑤ $20\sqrt[3]{7} \times 20\sqrt[3]{7} \times 10\sqrt[3]{7}$ cm

④ $(0, \pm\sqrt{2}, 0)$